Database Design – Course Project

Library Mobile Application Backend System

Role: Team Leader (Database Lead)
Inha University | Sept 2024 – Dec 2024

Project Overview

This project involved designing and implementing a **MySQL-based relational database system** for a library mobile application. The system supported barcode-based book identification, user management, and full CRUD (Create, Read, Update, Delete) operations through a Flask backend API integrated with a React Native (Expo) mobile application.

Role and Key Contributions

As **Team Leader**, I was responsible for task delegation, database design, and backend data architecture.

- ✓ Designed the complete **MySQL relational database schema**, including tables, primary/foreign keys, indexing strategy, and relationship constraints.
- ✓ Modeled entity relationships for users, books, loans, and barcode-based identification.
- ✓ Defined SQL data types and constraints to ensure data integrity and normalization (3NF).
- ✓ Implemented and tested full **CRUD operations** using SQL queries.
- ✓ Integrated RESTful endpoints to handle book search, borrowing records, and user authentication.
- ✓ Assigned development roles across frontend (React Native) and backend tasks.
- ✓ Delegated system testing and integration debugging to a designated error-handling member while focusing primarily on database architecture and API logic.

Technical Challenges and Solutions

1. Database Normalization and Relationship Design

Challenge:
Avoiding redundancy while maintaining efficient query performance for book and loan records.

Solution:

- Applied normalization principles (up to 3NF).
- Implemented foreign key constraints to enforce referential integrity.
- Used indexing on frequently queried fields (e.g., barcode, user ID).

---

2. Barcode-Based Book Identification

Challenge:
Ensuring each book copy could be uniquely identified and retrieved efficiently.

Solution:

- Designed a unique barcode field with indexed lookup capability.
- Structured queries to support fast book search and availability checks.

Outcome

The project resulted in a functional backend system supporting:

- ✓ Structured relational database design
- ✓ Barcode-based book tracking
- ✓ Secure and reliable CRUD operations
- ✓ Mobile app integration via Flask API

This project strengthened practical knowledge in **database schema design, relational modeling, backend API development, and team coordination**.